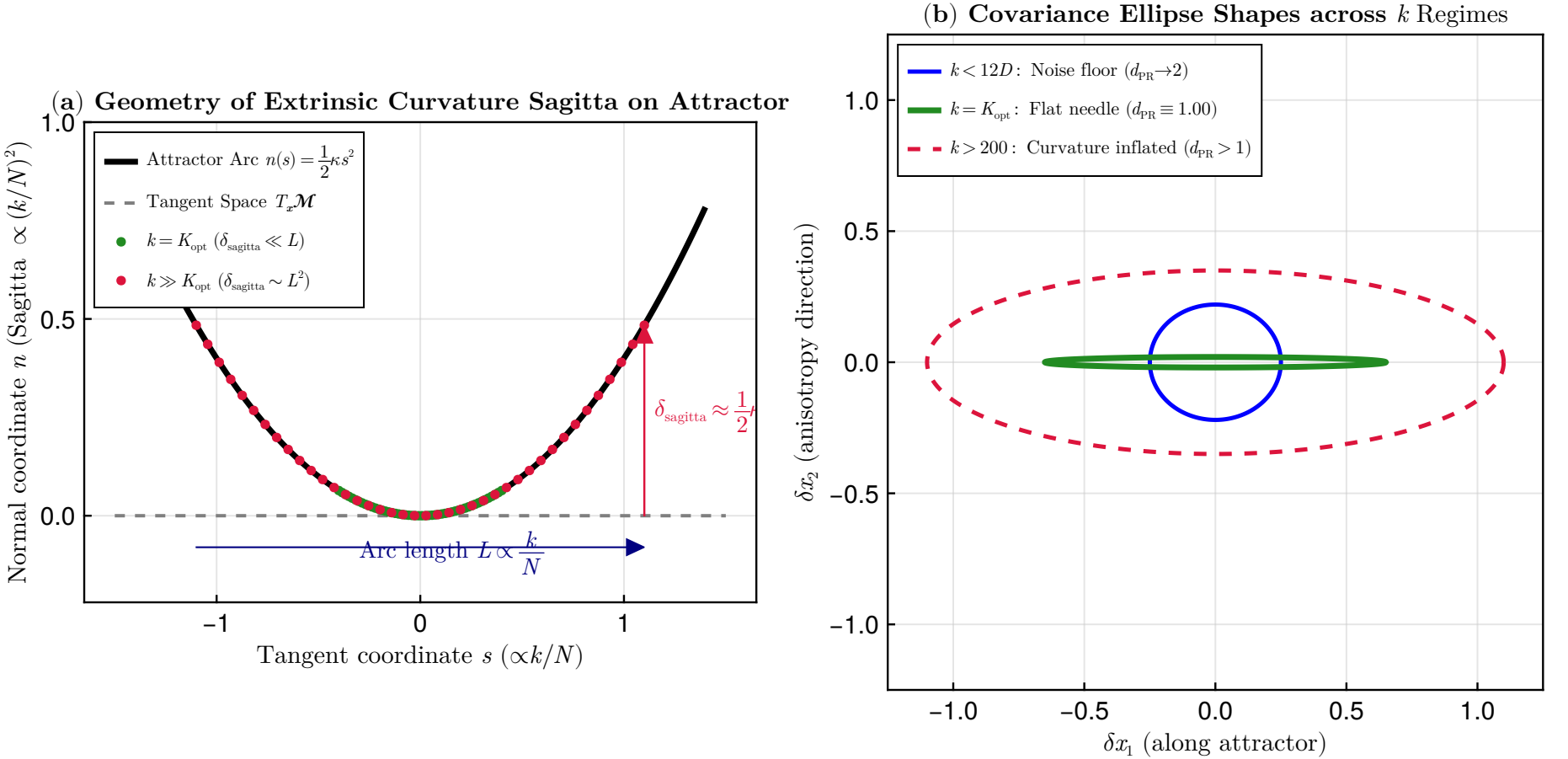
(a) Geometry of Extrinsic Curvature Sagitta on Attractor
Attractor Arc $n(s) = \frac{1}{2}\kappa s^2$
Tangent Space $T_x\mathcal{M}$
$k = K_{\text{opt}}$ ($\delta_{\text{sagitta}} \ll L$)
$k \gg K_{\text{opt}}$ ($\delta_{\text{sagitta}} \sim L^2$)
$\delta_{\text{sagitta}} \approx \frac{1}{2}\kappa$
Arc length $L \propto \frac{k}{N}$
Normal coordinate n (Sagitta $\propto (k/N)^2$)
Tangent coordinate s ($\propto k/N$)
1.0
0.5
0.0
−1
0
1
(b) Covariance Ellipse Shapes across k Regimes
$k < 12D$: Noise floor ($d_{\text{PR}} \to 2$)
$k = K_{\text{opt}}$: Flat needle ($d_{\text{PR}} \equiv 1.00$)
$k > 200$: Curvature inflated ($d_{\text{PR}} > 1$)
δx_2 (anisotropy direction)
δx_1 (along attractor)
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0